CD8 T Cell Fates

& Tumor Immunotherapy

Effectors · Memory · Exhaustion · CD4 Help · Checkpoint Immunotherapy

Killer Effector **Memory Cell** **Exhausted Cell**

10 Slides — Recitation 7

Three CD8 T Cell Fates: Overview

Killer Effector

- Immediate cytotoxic function — kills within hours
- Deploys Granzyme B, Perforin, TRAIL
- Secretes IFN-γ, TNF-α at high levels
- Expresses T-bet hi / Eomes lo
- Short-lived: undergoes clonal contraction
- ~90% die by day 7–10 post-infection
- Cannot self-renew; no homeostatic proliferation

Memory Cell

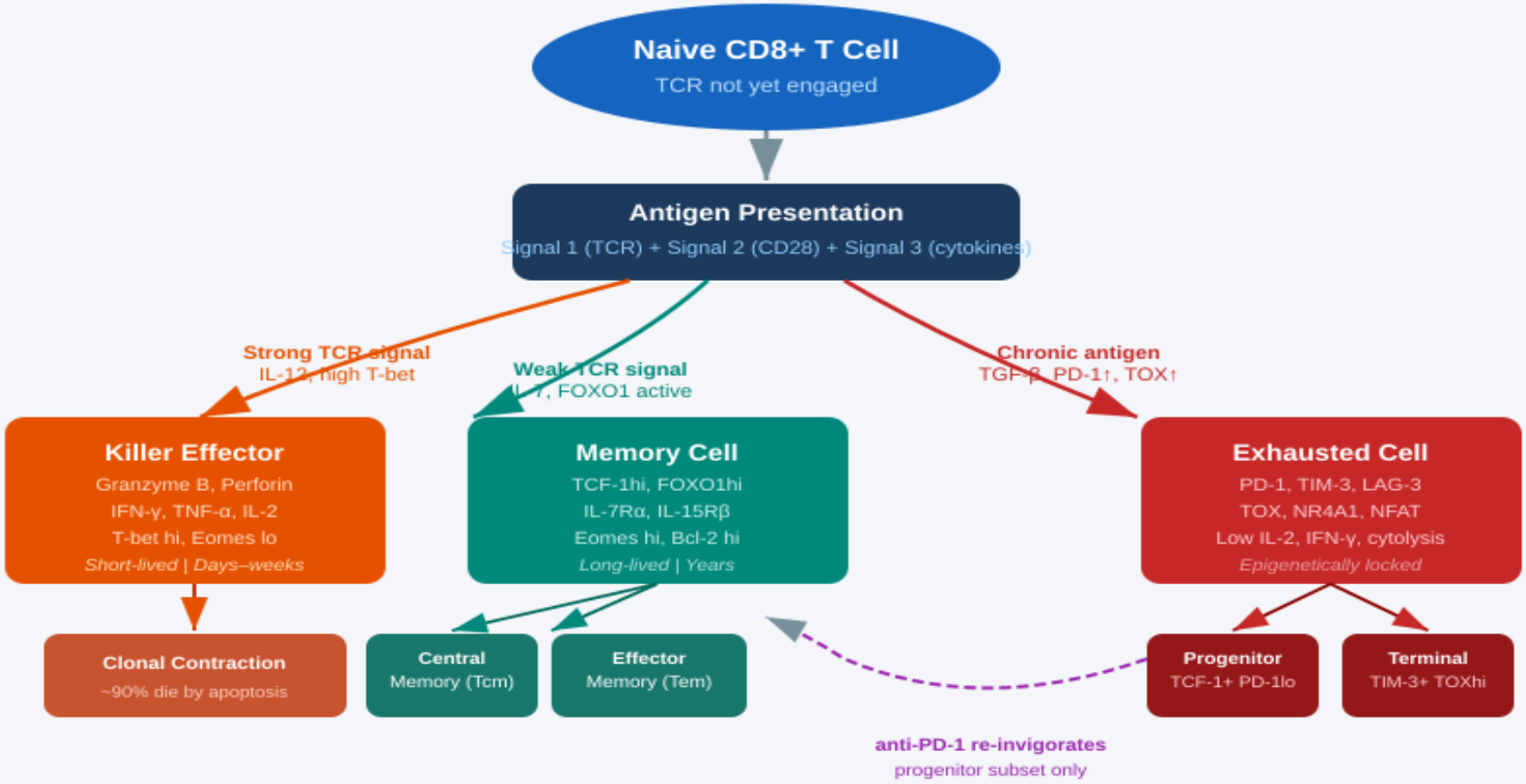
- Long-lived: persists for years/decades
- Expresses IL-7Rα (CD127) for survival
- High TCF-1 (Tcf7) for self-renewal
- Eomes hi, Bcl-2 hi (anti-apoptotic)
- Central (Tcm: lymph nodes) & Effector (Tem: tissue) subtypes
- Rapid recall on re-challenge (24–48 hrs)
- Maintained by homeostatic IL-7/IL-15 signaling

Exhausted Cell

- Progressive loss of function over time
- Hierarchical loss: IL-2 first → TNF → IFN-γ → CTL
- Co-inhibitory receptors: PD-1, TIM-3, LAG-3, TIGIT
- TOX transcription factor drives epigenetic lock
- Progenitor (TCF-1+ PD-1lo) → Terminal (TIM-3+ TOXhi)
- Metabolic dysfunction: impaired OXPHOS & glycolysis
- Partially reinvigorated by anti-PD-1 therapy

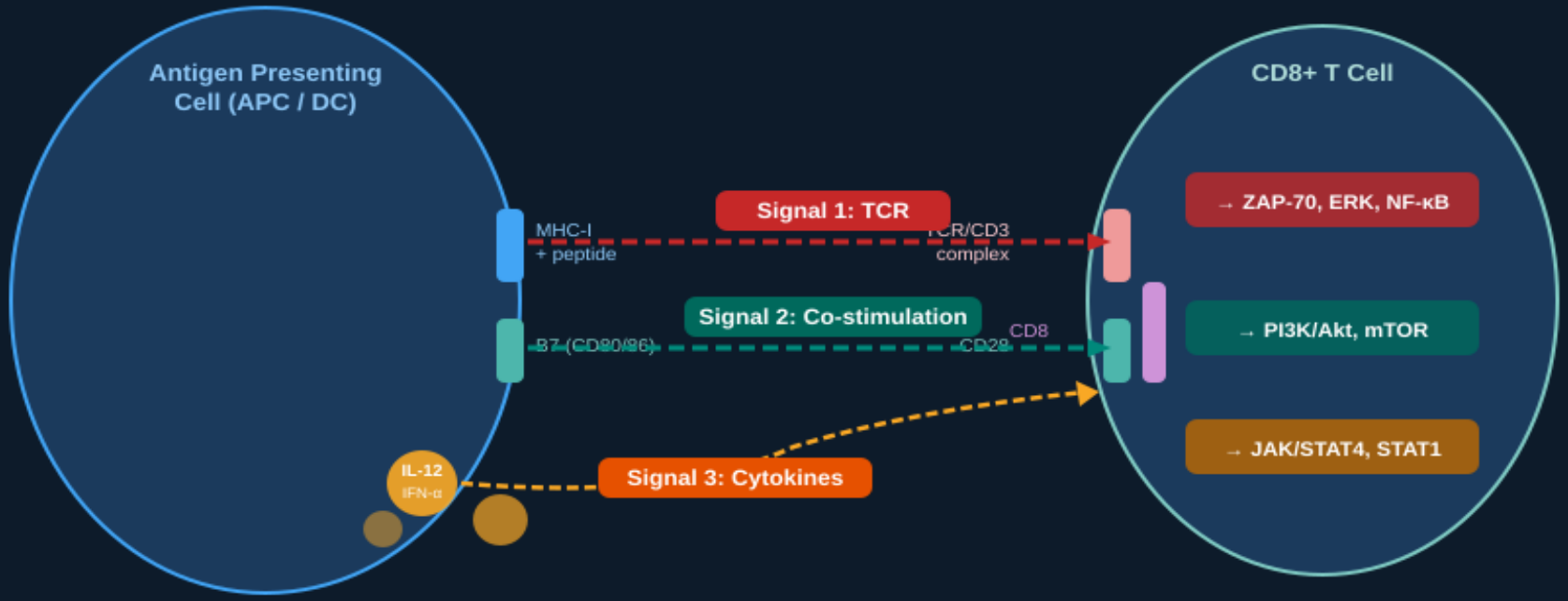
Interconversion: Effectors → irreversible. Progenitor Tex → partial re-invigoration with checkpoint blockade. Memory → re-activates on re-challenge.

CD8 T Cell Fate Decision Tree



CD8 T Cell Fate Decision Tree — Antigen-Driven Outcomes

Three Signals of T Cell Activation



All 3 signals
Full activation → Effector
+ Memory potential

Signal 1 only
Anergy / tolerance
No proliferation

Chronic Signal 1
Persistent stimulation
→ Exhaustion

Signals 1+2, weak
Low-affinity priming
→ Memory bias

Three-Signal Model of T Cell Activation and Fate

Conditions During Antigen Presentation That Bias Fate

1

TCR Signal Strength

- Low affinity → memory bias (FOXO1 active, TCF-1 preserved)
- High affinity → effector bias (mTOR active, T-bet↑)
- Chronic/persistent → exhaustion (TOX induction, PD-1↑)
- Signal strength measured by pMHC dwell time & CD8 co-receptor engagement

2

Co-stimulation (Signal 2)

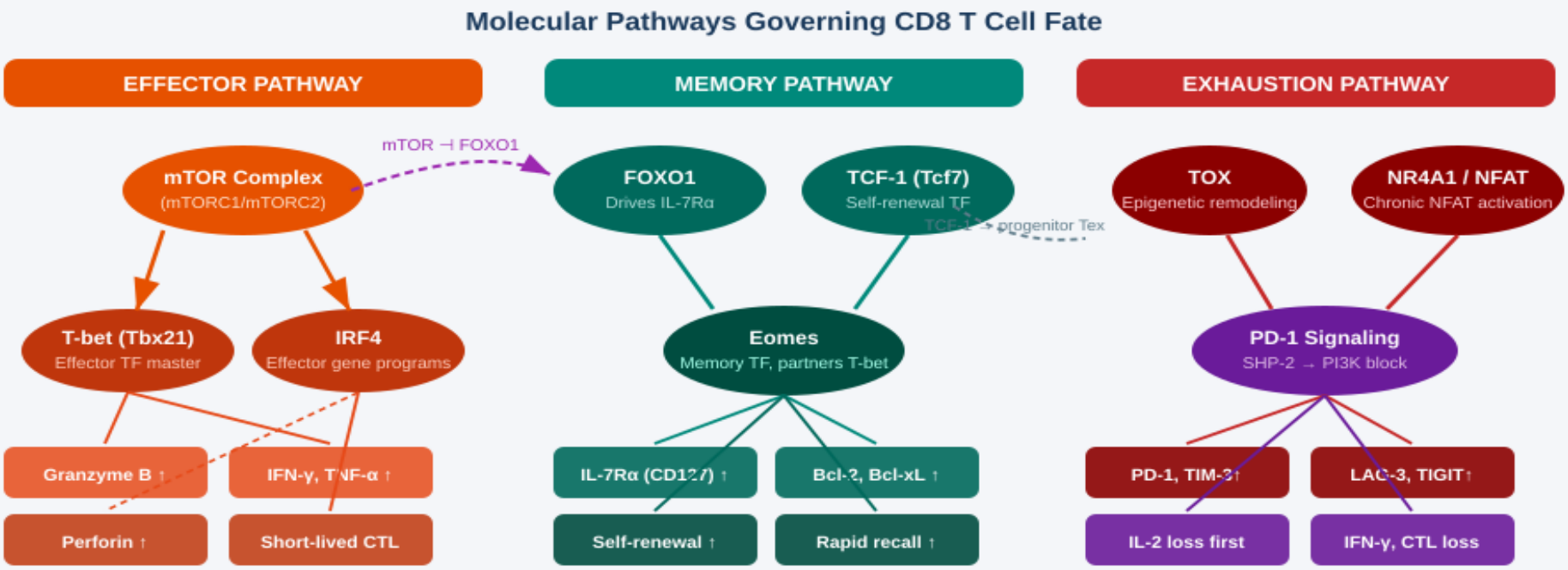
- CD28:B7 engagement activates PI3K → Akt → mTOR
- Strong CD28 → promotes both effector AND memory
- Weak/absent co-stim → anergy (NFAT without AP-1)
- CTLA-4 competes with CD28 for B7 binding (inhibitory)

3

Cytokine Milieu (Signal 3)

- IL-12 + IFN-α: strong T-bet induction → effector fate
- IL-2: survival signal; promotes effector + early memory
- IL-7/IL-15: homeostatic memory maintenance
- TGF-β + IL-10: exhaustion-promoting; SMAD signaling

Molecular Pathways: T-bet, FOXO1, TCF-1, TOX



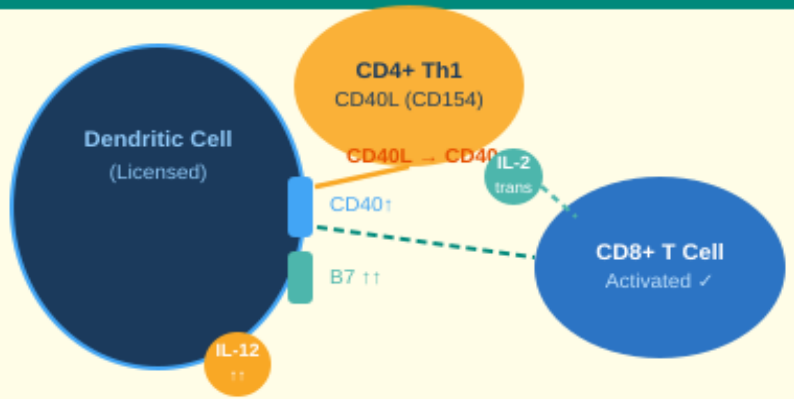
Key: Effector TFs / factors · Memory TFs / factors · Exhaustion TFs / factors · Cross-pathway inhibition

Note: T-bet (effector) and Eomes (memory) exist on a spectrum — high T-bet/low Eomes = short-lived CTL; low T-bet/high Eomes = memory. mTOR activation promotes T-bet via glycolytic metabolism. FOXO1 is phosphorylated (inactivated) by AKT/mTOR, linking nutrient sensing to cell fate. TOX is induced by persistent NFAT signaling and drives chromatin remodeling that stabilizes the exhaustion epigenome — making it difficult to reverse even after antigen removal.

Role of CD4 T Cell Help in CD8 Priming

Mechanism of CD4 T Cell Help for CD8 T Cell Priming

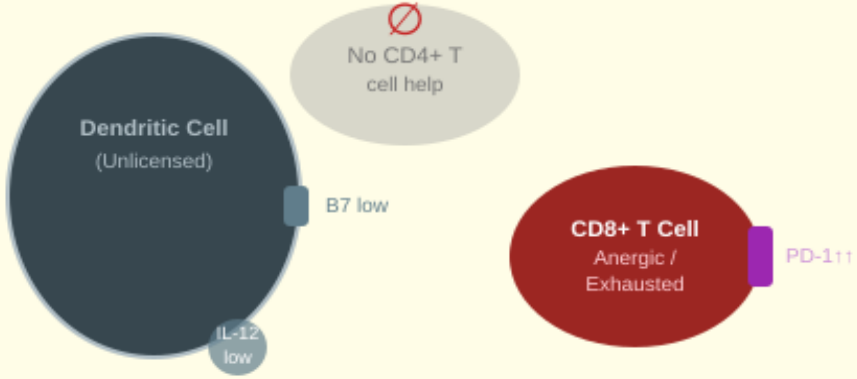
✓ WITH CD4 Help



Outcomes with CD4 Help:

- Strong CD8 clonal expansion (10–100×)
- Robust memory formation (TCF-1+ cells)
- Anti-apoptotic signals (Bcl-2 ↑)
- Prevention of exhaustion (PD-1 kept low)
- Functional recall upon re-challenge

✗ WITHOUT CD4 Help ("Helpless" Priming)



Outcomes without CD4 Help:

- Poor clonal expansion (limited proliferation)
- Failure to form functional memory
- Progressive PD-1 upregulation
- Susceptibility to AICD / deletion
- Tumor immune evasion facilitated

CD4 Help: Mechanisms & Vaccination Strategy

How CD4 Help Works

DC Licensing via CD40L→CD40

CD4 cells express CD40L which engages CD40 on DCs. This upregulates B7 co-stimulatory molecules and IL-12 production — directly improving the priming environment for CD8 T cells.

IL-2 Provision in trans

CD4 Th1 cells secrete IL-2, which acts in trans on nearby CD8 T cells. IL-2 drives proliferation, prevents activation-induced cell death (AICD), and promotes memory precursor formation.

Prevention of Exhaustion

CD4 help sustains optimal co-stimulatory signals. Without CD4, DCs fail to upregulate B7, leading to sub-threshold CD28 signaling and eventual PD-1-driven exhaustion of CD8 cells.

Treg Suppression of Help (in tumors)

In the tumor microenvironment, Foxp3+ Tregs deplete IL-2 and suppress DC function — effectively removing CD4 help and leaving CD8 cells vulnerable to exhaustion.

Vaccination Strategy Implications

Include CD4 Epitopes (MHC-II)

Vaccines must contain peptides that bind MHC class II to activate CD4 T cells. Without this, CD8 priming may be quantitatively reduced and qualitatively inferior.

Adjuvants That Activate Innate Immunity

TLR agonists (poly-ICLC, CpG, MPLA) boost DC maturation, IL-12, and type I IFN production — amplifying both Signal 2 and Signal 3 to support CD8 memory.

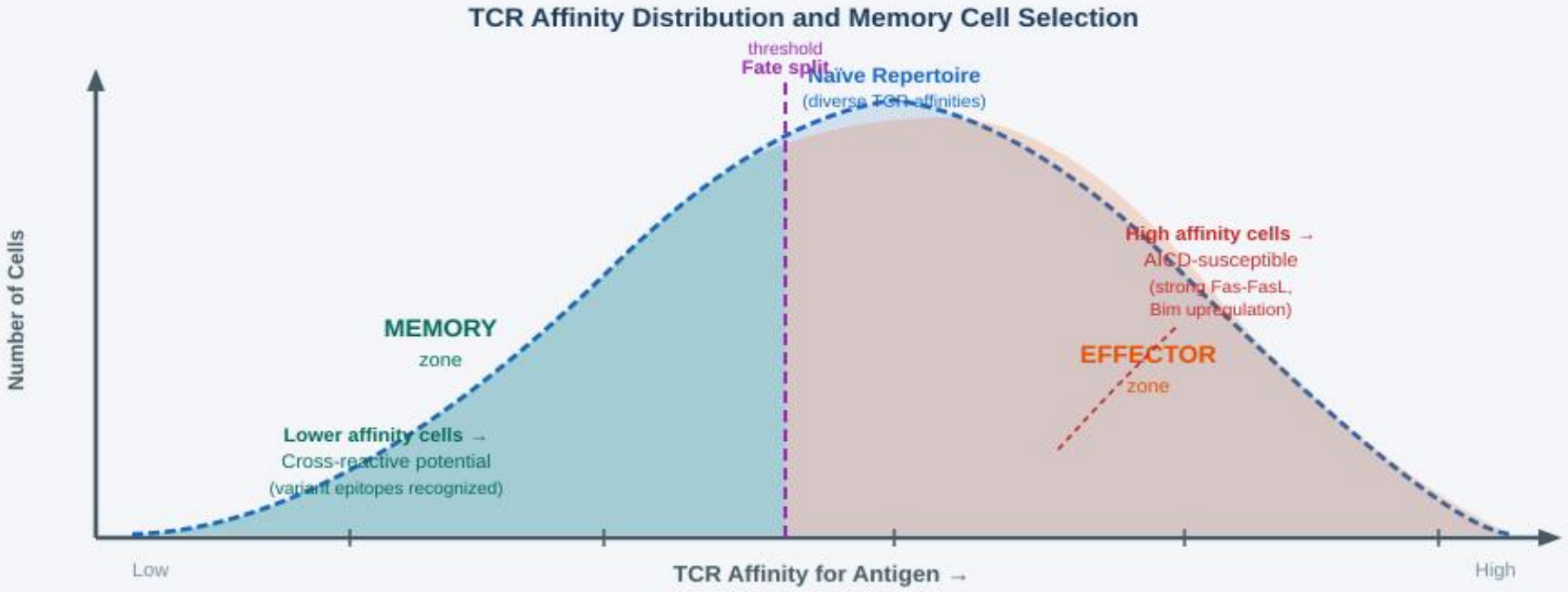
mRNA / Lipid Nanoparticle Vaccines

mRNA vaccines encoding tumor neoantigens can present both MHC-I and MHC-II epitopes simultaneously, providing integrated CD4 and CD8 help from a single immunogen.

Timing & Antigen Persistence

Prolonged antigen depots (slow-release adjuvants) can improve memory formation but excessive persistence risks exhaustion — balancing is critical in tumor vaccines.

TCR Repertoire, Affinity, and Memory Cell Selection



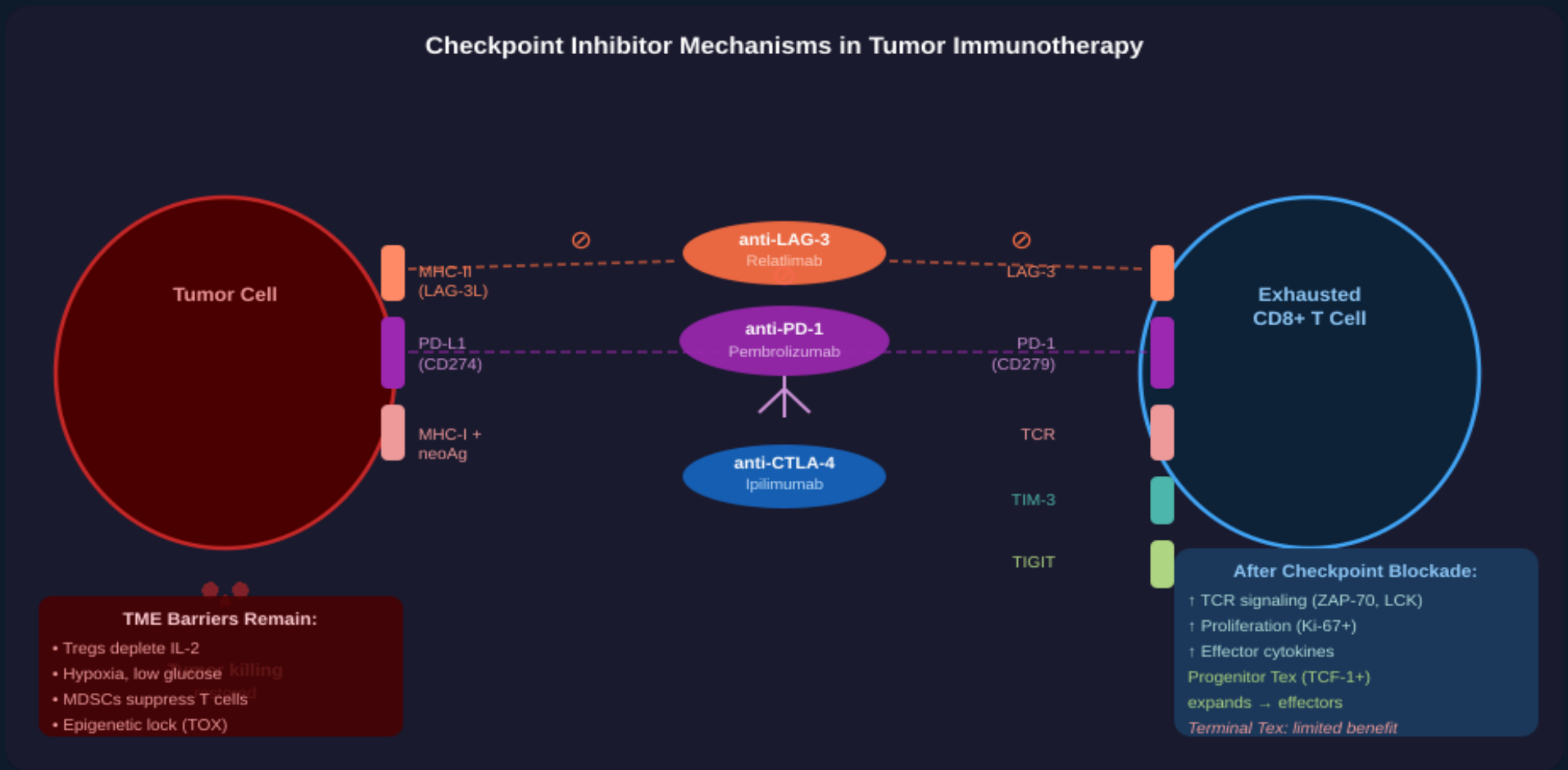
Breadth of Coverage
Low-affinity clones cover more epitope variants & drifted pathogens (tumor neoantigens)

Population Resilience
Diverse memory pool ensures that no single escape mutation eliminates ALL protective clones

Recall Sensitivity
Memory TCRs tuned for moderate affinity: sensitive enough to detect re-infection without auto-reactivity

Avoidance of AICD
High-affinity cells upregulate Fas, Bim during peak response → die during contraction
Low-affinity clones survive

Tumor Immunotherapy — Checkpoint Inhibitor Mechanisms



T Cell Exhaustion — Drivers, Stages, and Therapeutic Targets

STAGES OF EXHAUSTION

Effector-like

Early, partial function
PD-1lo, TCF-1+, proliferative
still responds to stimulation

Progenitor Exhausted

TCF-1+, PD-1int, TOXlo
Self-renewing subset
Responds to anti-PD-1 therapy

Transitory

Ki-67+, expanding
Generated from progenitors
after checkpoint blockade

Terminal Exhausted

TIM-3+, TOXhi, CX3CR1-
No self-renewal, AICD prone
Poor response to therapy

THERAPEUTIC TARGETS

anti-PD-1 / anti-PD-L1

Pembrolizumab, Nivolumab, Atezolizumab

Blocks SHP-2 activation; restores TCR signaling in progenitor Tex. Most effective at progenitor stage.

anti-CTLA-4

Ipilimumab, Tremelimumab

Acts at priming phase. Depletes Tregs in TME via ADCC. Synergizes with anti-PD-1 in combination therapy.

anti-LAG-3

Relatlimab (+ Nivolumab = Opdualag)

LAG-3 binds MHC-II; blockade enhances TCR signaling. Approved for melanoma in combination with PD-1.

anti-TIM-3 / anti-TIGIT

Sabatolimab, Tiragolumab (trials)

Target terminally exhausted cells. In trials alone and in combination. TIGIT competes with CD226 for CD155.

Key Takeaways

A Three fates arise after antigen exposure: Effectors kill acutely; Memory cells self-renew for durable protection; Exhausted cells are epigenetically locked into dysfunction — but progenitor subsets remain targetable.

B Fate is determined by TCR signal strength, co-stimulation (CD28:B7), and cytokine context (IL-12 vs. TGF-β). Key TFs: T-bet (effector), FOXO1/TCF-1/Eomes (memory), TOX/NR4A1 (exhaustion). mTOR vs. FOXO1 balance is central.

C CD4 T cell help licenses DCs (CD40L→CD40), provides IL-2 in trans, and prevents exhaustion. Without it, CD8 cells undergo helpless priming. Vaccines must include MHC-II epitopes and Th1-promoting adjuvants.

D Lower-affinity TCR clones survive contraction to form memory (avoiding AICD via weaker Fas/Bim signals). This preserves breadth and cross-reactivity — critical for variant pathogens and mutated tumor neoantigens.

"The identity of a T cell is written not just in its genes, but in the history of its encounters with antigen."